Episode-2

Sunday, 4 October 2026 12:45 PM

SCISOFT ACADEMY

OBJECT ORIENTED PROGRAMMING - EP-②

CLASS & OBJECT

What we are going to cover ?

1) Recap: Object = Data + Behaviour
2) What is class ?
3) What is an Object?
4) Class vs Object
5) Attributes / Properties
6) Methods
7) Creating objects
8) Multiple objects
9) Each object has its own state
10) Objects interaction
11) First simple Code
12) Real-world Example

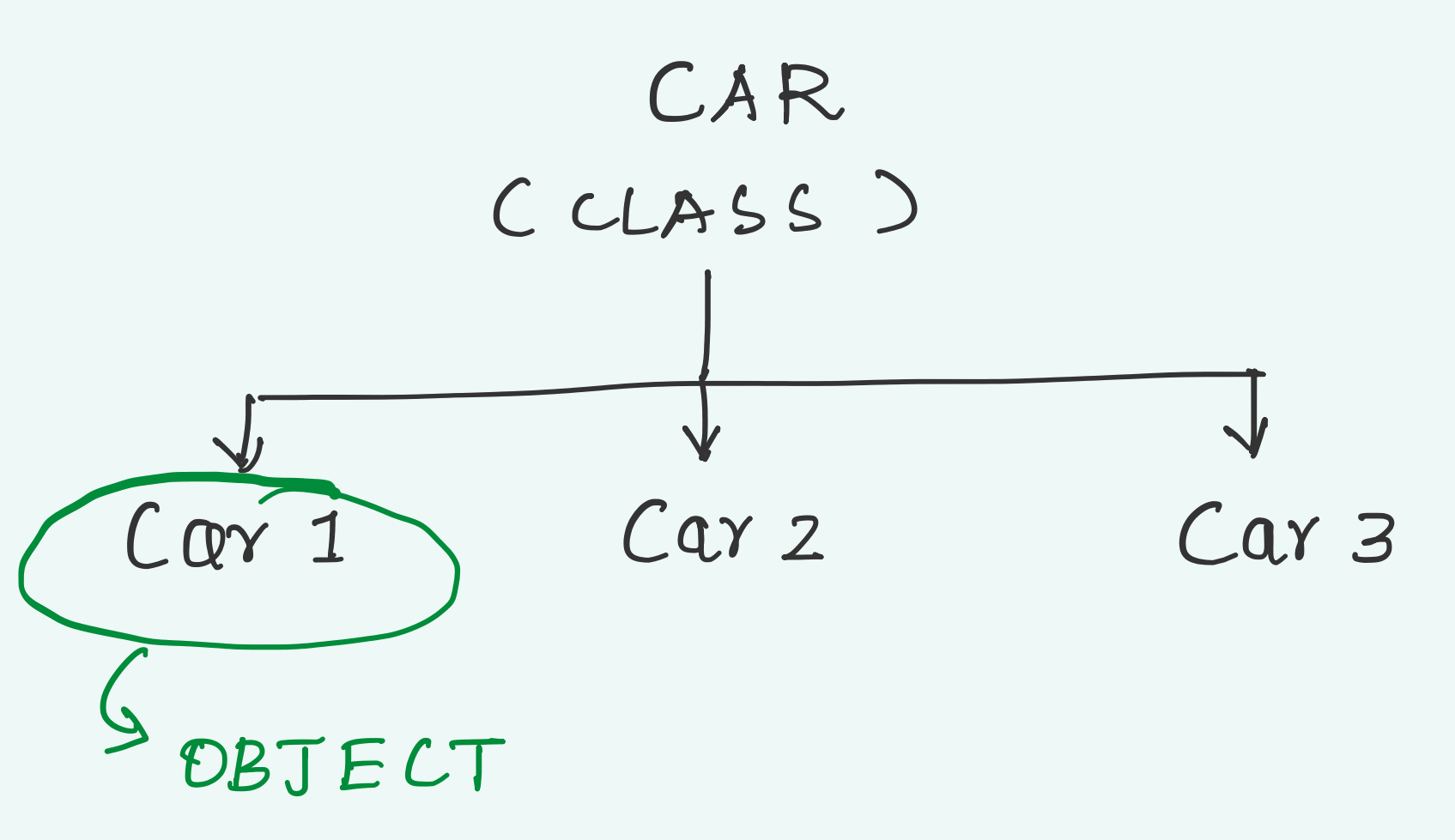
CLASS ≠ OBJECT

CAR

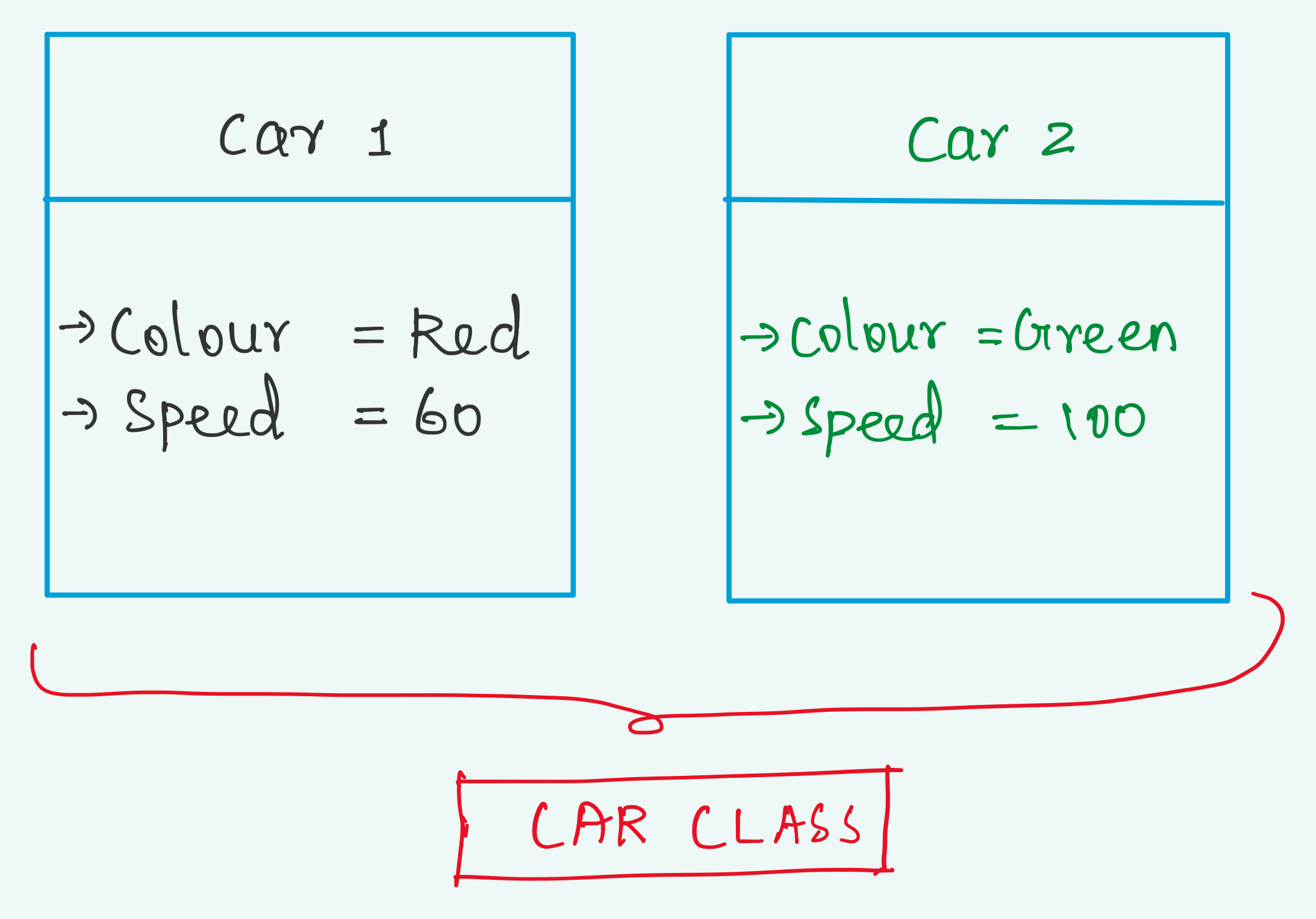
→ Colour
→ Speed
→ Fuel

→ start ()
→ Accelerate ()
→ Brake ()

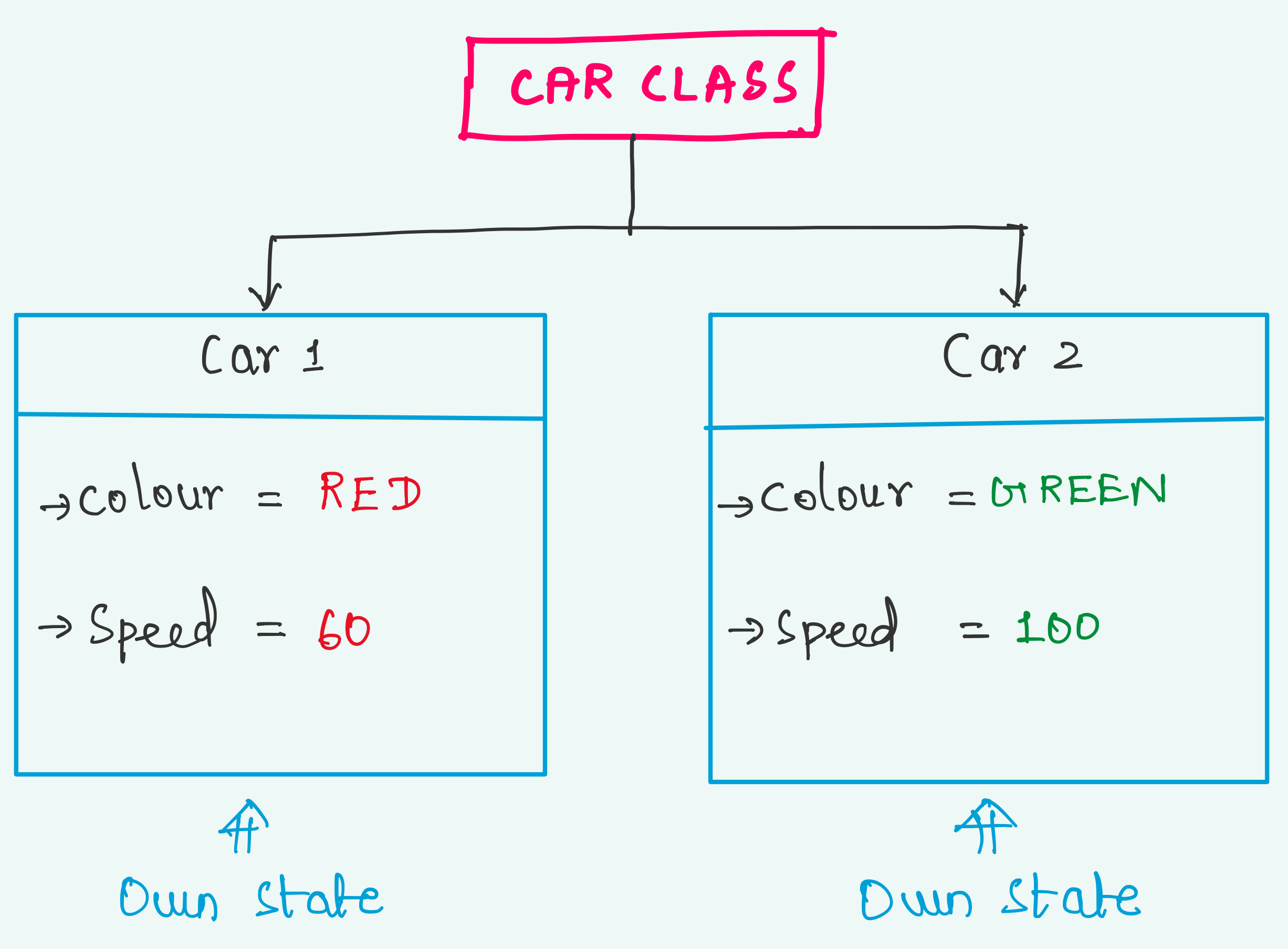
CLASS
↓
Blueprint / Definition


CAR
(CLASS)
Car 1
Car 2
Car 3
OBJECT

*** Each object has separate state**


Car 1
→ Colour = Red
→ Speed = 60
Car 2
→ colour = Green
→ speed = 100
CAR CLASS

* Both states same ?? ⇒ NO


CAR CLASS
Car 1
→ colour = RED
→ Speed = 60
Car 2
→ colour = GREEN
→ Speed = 100
Own State
Own State

ATTRIBUTES & METHODS

Attributes	Methods
① colour	① start ()
② Speed	② accelerate ()
③ Fuel	③ brake ()

Represents the variables of the objects — [Attributes / Fields / Properties]

Represents the behaviour of the objects — [Methods / Functions]

Language - neutral Pseudocode:

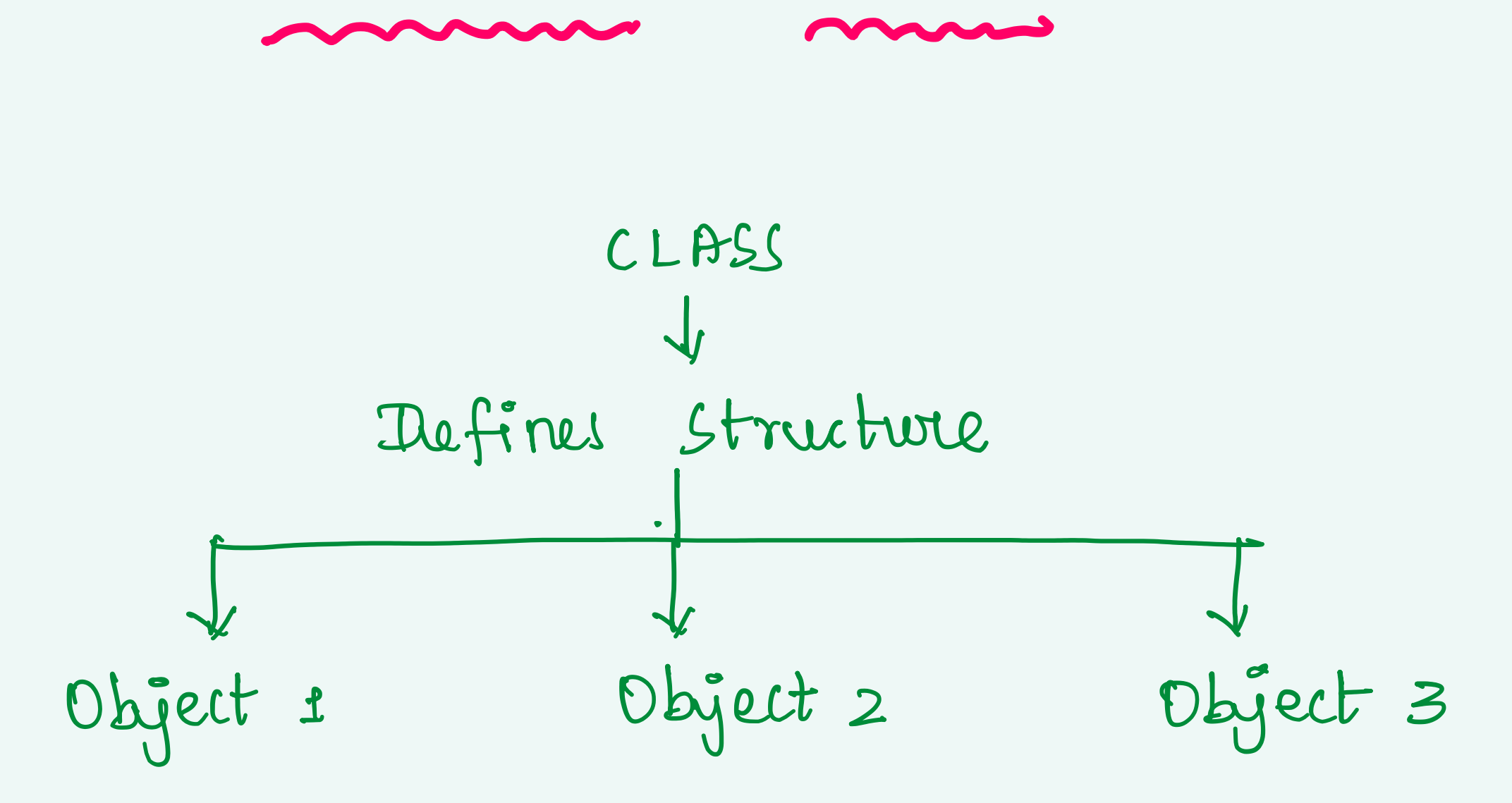
```
 class Car
{
    colour
    speed
    fuel

    start ()
    accelerate ()
    brake ()
 }
```

⇒ car1 = Car ()
⇒ car2 = Car ()
} ⇒ Instantiation

Car ()
↓
Object Created
↓
car1

IMPORTANT NOTES

①


CLASS
Defines Structure
Object 1
Object 2
Object 3

② CLASS → Blueprint / Definition

③ OBJECT → Actual Instance

④ STATE → Data

⑤ METHOD → Behaviour

NEXT VIDEO

How actual object initializes when Car () is called ?